

Multi-Agent AI System for Risk-Aware, Explainable Hydrocarbon

HACKATHON

Exploration

AI & HYDROCARBON

Accelerating Discovery with Trustworthy, ESG-Compliant AI

EXPLORATION

EW Hackathon 2025, IIGIPT

TEAM

Prathamesh Dhote | Final Year, IT (22IT031)

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INDIA ENERGY WEEK

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High-Stakes Decisions: The Need for Trustworthy AI

FINANCIAL RISK

Each exploration well costs **\$50M – \$500M**, yet only **~30%** succeed. Non-productive drilling represents massive financial and environmental loss.

DATA HETEROGENEITY

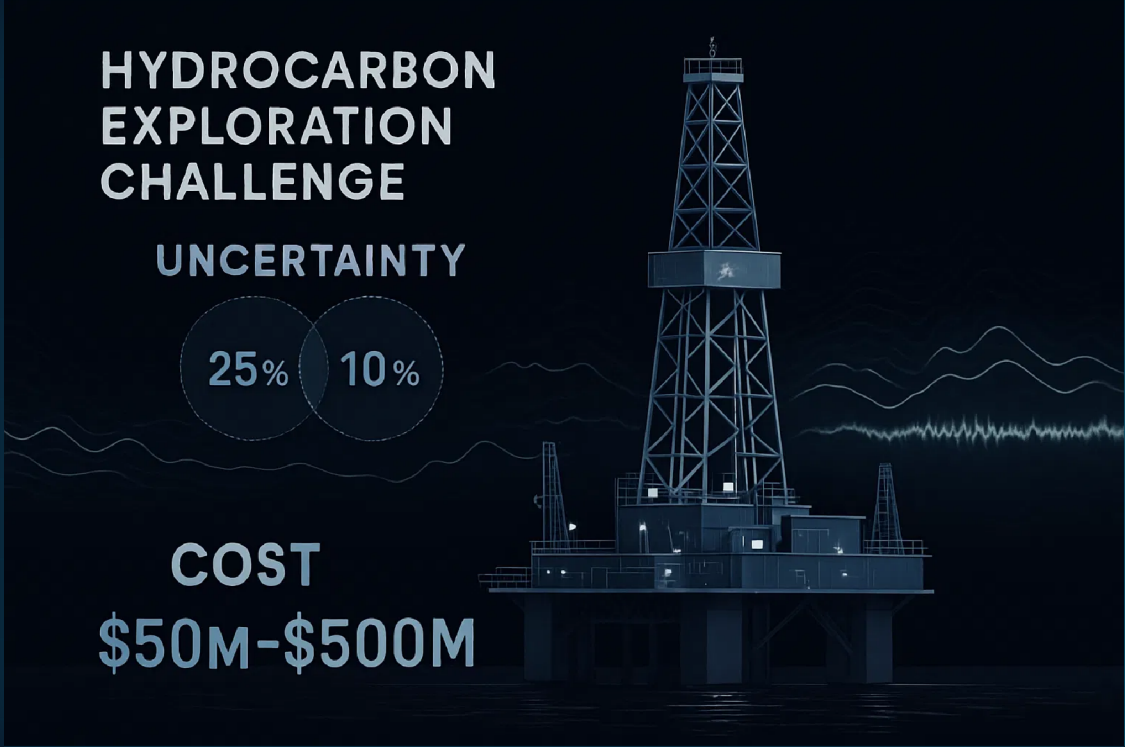
Exploration involves massive, multi-modal data: 3D seismic, well logs, economic metrics, and ESG constraints. Traditional single-model approaches fail to integrate this complexity.

BLACK BOX PROBLEM

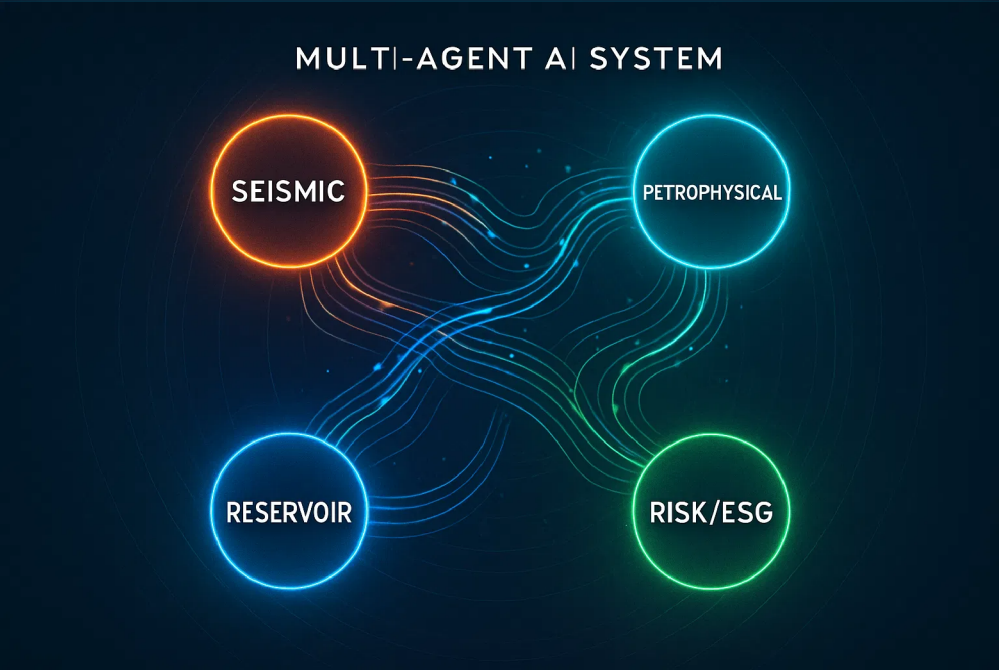
Current ML models lack transparency, interpretability, and uncertainty quantification, leading to low trust from geoscientists and decision-makers.

REGULATORY PRESSURE

Increasing demand for auditable, risk-aware, and environmentally compliant exploration decisions. ESG compliance is now non-negotiable.



Introducing AEIS: A Multi-Agent System for Integrated Decision-Making



MULTI-AGENT SPECIALIZATION

AEIS is a collaborative system of **four specialized AI agents**, each optimized for a specific domain: Seismic, Petrophysical, Reservoir, and Risk/ESG.

EXPLAINABILITY BY DESIGN

Every prediction includes **robust uncertainty quantification** (Monte Carlo Dropout) and clear rationale (SHAP values, scenario analysis).

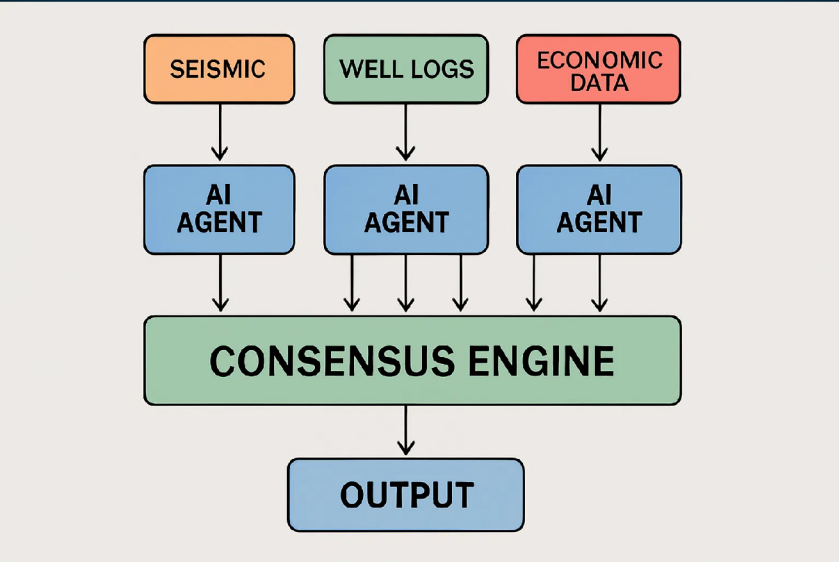
RISK-AWARE CONSENSUS

A central Consensus Engine fuses agent predictions based on confidence, with a critical **Hard Veto** from the Risk/ESG agent for non-compliant locations.

VALUE PROPOSITION

Higher drilling success, reduced non-productive time, faster decision cycles (months to hours/days), and end-to-end compliance.

From Raw Data to Auditable Recommendation: The AEIS Workflow



AGENT	CORE FUNCTION	KEY MODEL ARCHITECTURE
1. Seismic	Fault & Horizon Detection	CNN + Self-Attention
2. Petrophysical	Lithofacies Classification (6 classes)	Hybrid 1D CNN + XGBoost Ensemble
3. Reservoir	Hydrocarbon Potential & Quality Index	Deep Residual CNN + MLP (Physics-Informed)
4. Risk/ESG	Economic Viability & Compliance Veto	Bayesian Monte Carlo + Rule Engine

Agent 2: Robust Lithofacies Classification

PETROPHYSICAL INTELLIGENCE WITH UNCERTAINTY

DATASET & SCALE

FORCE 2020 (Norwegian Shelf)

118 wells | ~75k windows (50-sample, 50% overlap)

10 log curves: GR, NPHI, RHOB, DTC, DTS, RDEP, RMED, PEF, CALI, BS

6 lithology classes: Sandstone, Claystone, Chalk, Marl, Dolomite, Halite

PERFORMANCE METRICS

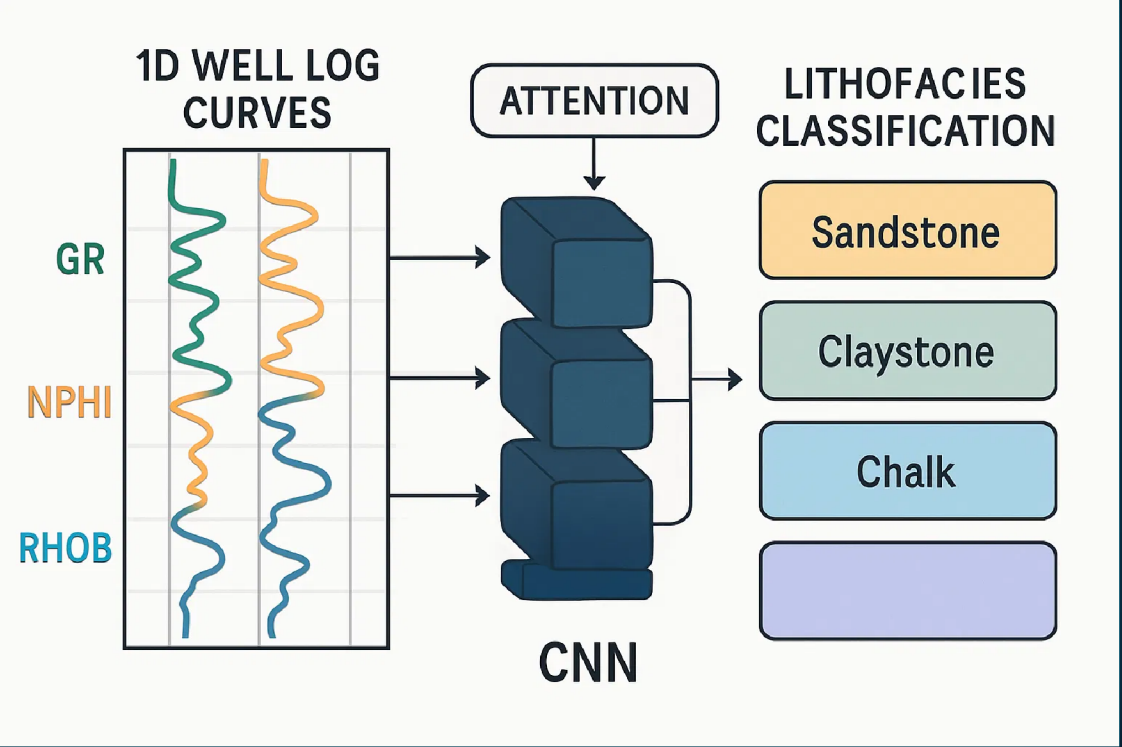
ACCURACY	F1 SCORE
85.2%	0.84

TOP CLASSES ACCURACY

Sandstone	89%
Halite	91%
Dolomite	86%

Key Innovation

Monte Carlo Dropout (20 iterations) provides explicit epistemic and aleatoric uncertainty for every lithology prediction, enabling confidence-aware decision-making.



Agent 3: Reservoir Prediction

PHYSICS-INFORMED DEEP LEARNING

OBJECTIVE

Predict Reservoir Quality Index (RQI) and Hydrocarbon Presence Probability by combining spatial feature extraction with petrophysical relationships anchored in reservoir physics.

MODEL ARCHITECTURE

- ▶ Deep Residual CNN for spatial feature extraction
- ▶ MLP Branch for petrophysical relationships
- ▶ Physics constraints (Darcy's law, Archie's equation)
- ▶ Monte Carlo Dropout for uncertainty quantification

PERFORMANCE METRICS

R² SCORE
0.91

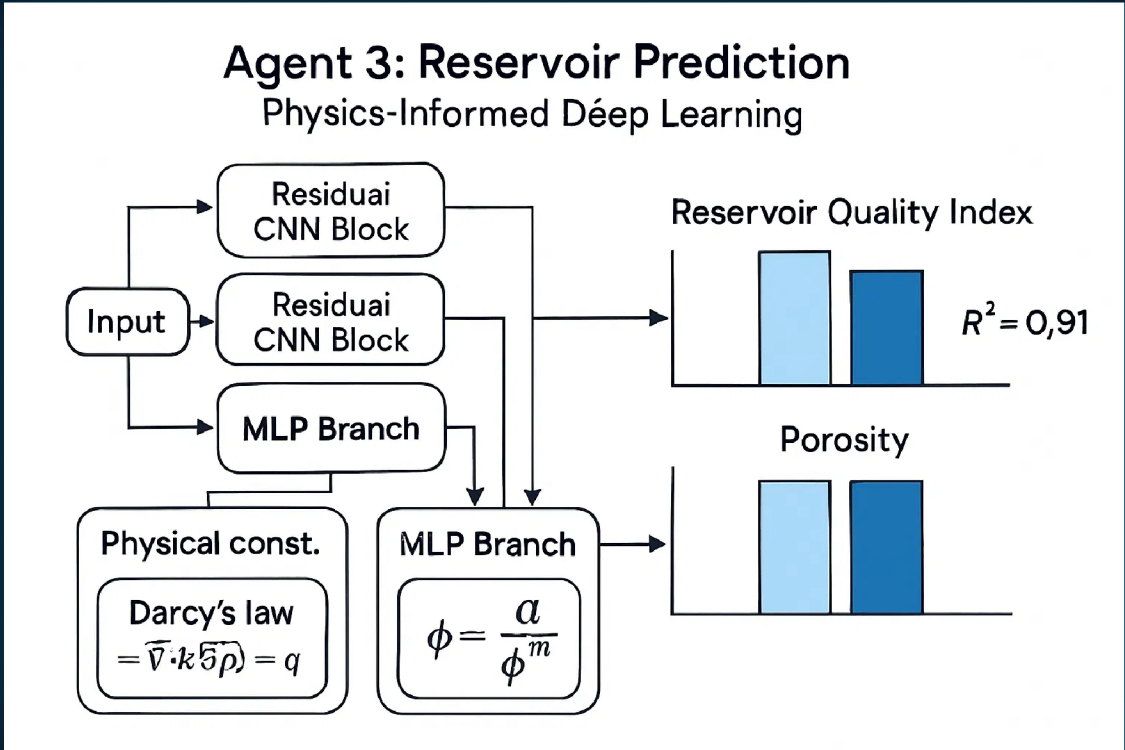
RMSE
0.028

MAE
0.021

VAL R²
0.92

PHYSICS INTEGRATION

Model is anchored by physical constraints ensuring predictions align with geoscientific reality. Top features: RHOB (32%), Lithology (18%), GR (14%).



Agent 4: Ensuring Financial and Environmental Compliance

ECONOMIC MODULE

Uses **Bayesian Monte Carlo Simulation** (10,000 scenarios) to forecast Net Present Value and Return on Investment with **P10/P50/P90** confidence intervals.

- ▶ NPV forecasting with probabilistic scenarios
- ▶ ROI calculation across uncertainty ranges
- ▶ Payback period analysis

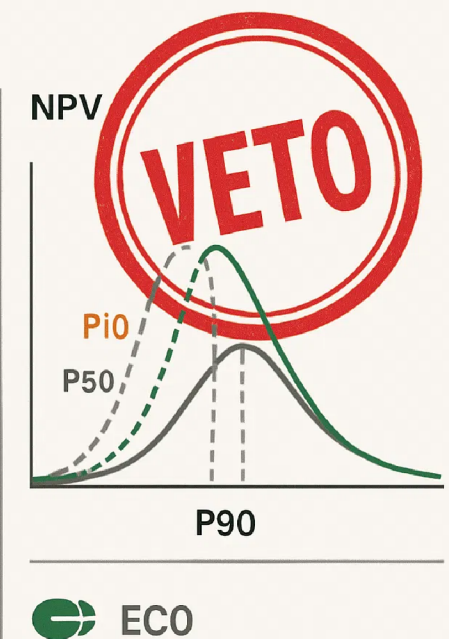
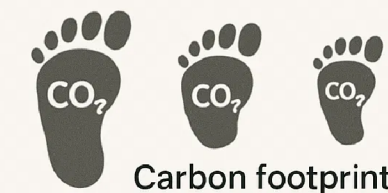
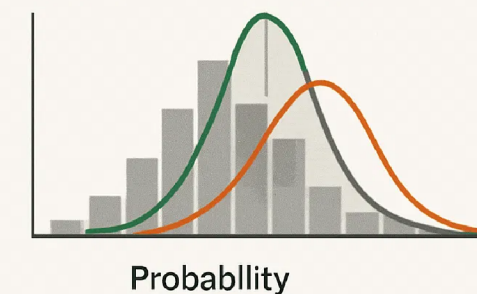
ESG MODULE

Calculates **carbon footprint** (kg CO₂/barrel) and water usage, checking against real-time regulatory compliance rules.

- ▶ Carbon emissions quantification
- ▶ Water consumption tracking
- ▶ Regulatory compliance scoring

Agent 4 (Risk/ESG)

Monte Carlo simulation



THE HARD VETO

If the ESG score falls below a predefined threshold, the agent issues a **Hard Veto**, overriding all technical and economic recommendations to

Trust and Transparency: Fusing Predictions and Providing Rationale

CONFIDENCE-WEIGHTED FUSION

The final recommendation is a **weighted average** of Agent 2, 3, and 4 scores, where the weight is inversely proportional to the agent's predicted uncertainty.

$w = 1/\sigma^2$ (weight inversely proportional to variance)

SHAP-BASED RATIONALE

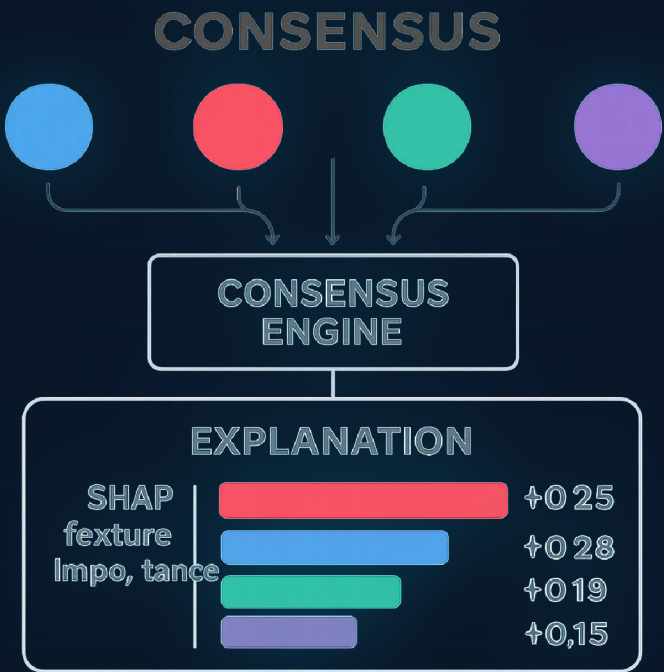
The system uses **SHAP (SHapley Additive exPlanations)** to attribute the final decision to the most influential input features, such as "High porosity (+0.35 impact)" or "Low drilling cost (+0.18 impact)".

SCENARIO ANALYTICS

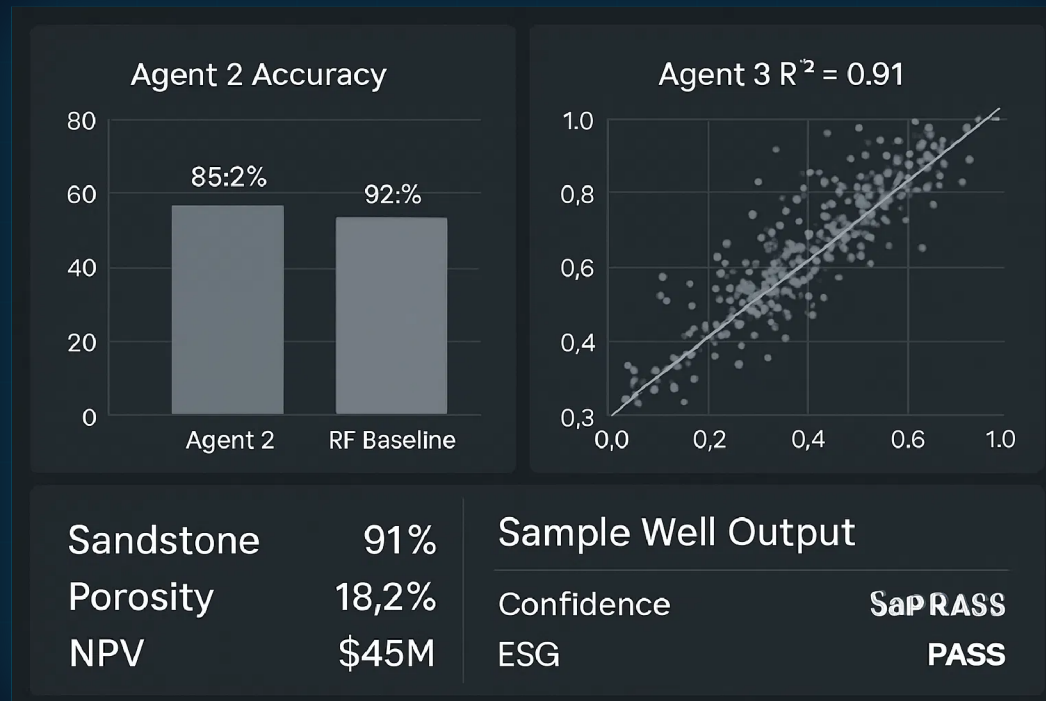
Provides **counterfactual analysis** for decision-makers: "If oil drops to \$60/bbl, NPV is \$28M (still viable)" or "If porosity <12%, REJECT".

AUDITABLE OUTPUT

A clear, **auditable report** showing the final decision, confidence level, risk assessment, and key drivers—enabling transparency and knowledge transfer.



Validated Performance: Higher Confidence, Lower Risk



ACCURACY UPLIFT

Agent 2 achieved **85.2% accuracy** in lithology classification, a **+3.2% improvement** over the Random Forest baseline (82%).

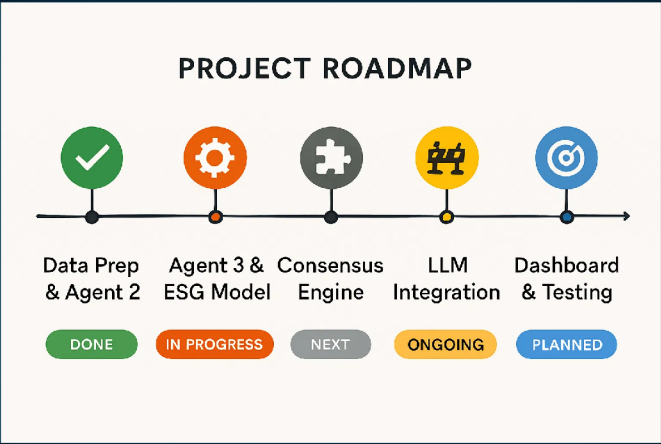
HIGH REGRESSION FIDELITY

Agent 3 demonstrated strong predictive power for reservoir quality with **$R^2 = 0.91$** , validating the physics-informed hybrid approach.

SAMPLE WELL OUTPUT

Well X at 2450m: Sandstone (91% conf.), Porosity 18.2%, NPV \$45M, ESG: **PASS** → **Recommend DRILL.**

Project Status: From Core Implementation to Full System Integration



PHASE	GOAL	STATUS	NEXT STEPS
Data Prep & Agent 2	Data Ingestion & Lithofacies Model	✓ DONE	SHAP Explainability (In Progress)
Agent 3 & ESG Model	Reservoir & Risk/ESG Model Development	⚙️ IN PROGRESS	Complete training & define Veto Rules
Consensus Engine	Multi-Agent Fusion & Weighted Voting	🧩 NEXT	Integrate Agent 2, 3, and 4 outputs
LLM Integration	Report Generation & Chatbot	🤖 ONGOING	Structure prompt for human-readable reports
Dashboard & Testing	Interactive UI & Robustness Testing	🎯 PLANNED	3D visualization & adversarial testing

Why AEIS is the Future of Exploration AI

COMPETITIVE ADVANTAGE



INTEGRATED
WORKFLOW



RISK
QUANTIFICATION



ESG-FIRST
APPROACH



AUDITABLE
& EXPLAINABLE

INTEGRATED WORKFLOW

Moves beyond **fragmented, single-model solutions** to a unified, end-to-end decision system that seamlessly integrates seismic, petrophysical, reservoir, and economic data into a cohesive framework.

RISK QUANTIFICATION

Explicitly models and reports **uncertainty at every step**, providing confidence intervals and probabilistic forecasts required by engineers and investors for informed decision-making.

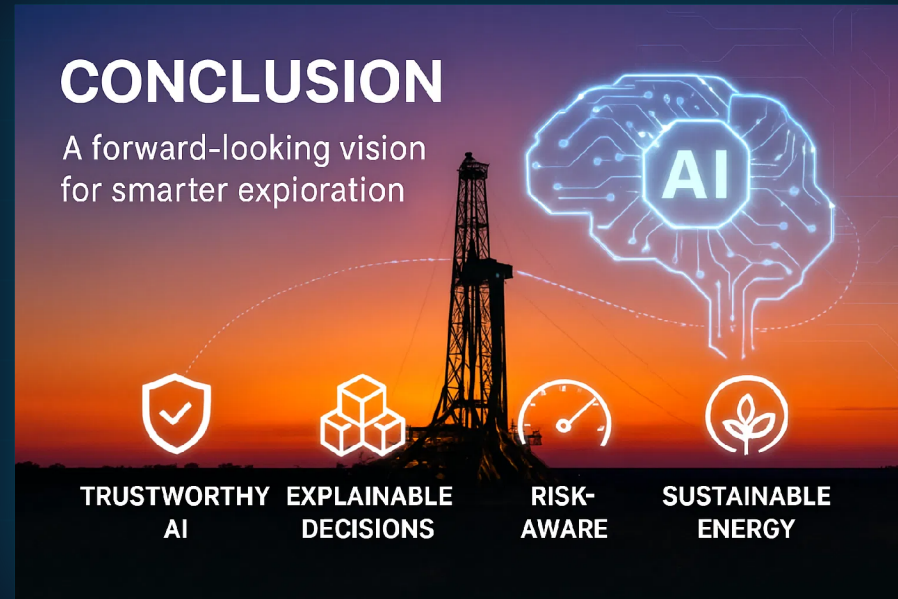
ESG-FIRST APPROACH

The mandatory **Hard Veto** ensures environmental and regulatory compliance is a non-negotiable part of the decision process, positioning AEIS as essential for responsible energy development.

AUDITABLE & EXPLAINABLE

Provides a clear, **SHAP-driven rationale** for every recommendation, building trust with stakeholders and facilitating knowledge transfer across technical and regulatory boundaries.

The Path to Smarter, Safer, and More Sustainable Exploration



AEIS delivers a **Trustworthy, Explainable, and Risk-Aware** solution to the high-stakes problem of hydrocarbon exploration. Our multi-agent architecture and consensus engine provide a robust framework for handling complex, heterogeneous data while integrating **ESG compliance** as a non-negotiable core principle.



TRUSTWORTHY AI



EXPLAINABLE DECISIONS



RISK-AWARE



SUSTAINABLE ENERGY

Call to Action: Enabling the Future of Responsible Energy



TRUSTWORTHY AI



EXPLAINABLE DECISIONS



RISK-AWARE



SUSTAINABLE ENERGY

NEXT STEPS & SUPPORT

- ▶ Finalize Agent 3 and integrate the Consensus Engine for full multi-agent deployment
- ▶ Deploy AEIS as a pilot project with industry partners to validate real-world performance
- ▶ Establish collaboration framework for knowledge transfer and continuous improvement
- ▶ Transform hydrocarbon exploration into a smarter, safer, and more responsible practice

Thank You

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